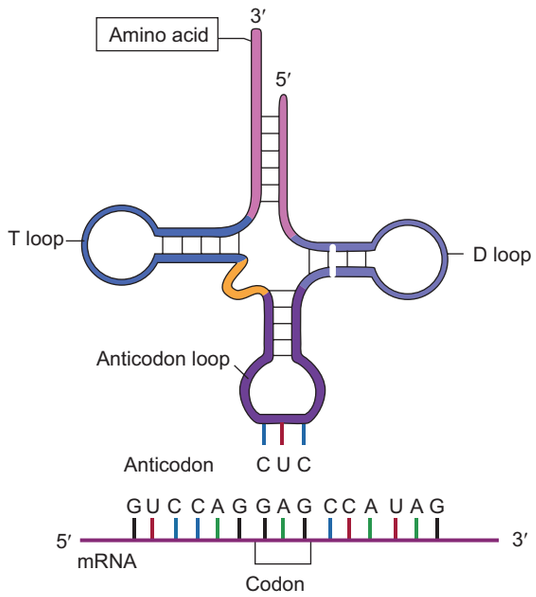


Figure 12.5: tRNA

tRNA is often designated **tRNA^{fMet}**. Both tRNA^{fMet} and tRNA^{Met} recognize the codon AUG, but only tRNA^{fMet} is used for initiation. All prokaryotic proteins while being synthesized have fMet at the amino terminus. However, this amino acid is frequently deformylated or removed later.

- The usual form of translation unit is a polyribosome or polysome wherein an mRNA is covered with ribosomes.

Redundancy and the Wobble Hypothesis

The identity of the third base of a codon appears to be unimportant. (The first base in a codon is at the 5′ end and the third base is at the 3′ end). Wobble refers to the less stringent requirement for base pairing at the third position of the codon than at the first two positions. That is the first two bases must follow Watson and Crick base pairing rule (A with U, or G with C), but the third base pair can be of a different type (for example, G with U). The Wobble hypothesis explains the pattern of redundancy in the code in that certain anticodons. (For example, those containing U and G in the first position of the anticodons) can pair with several codons during translation (Figure 12.6).

12.5 Types of Mutations

The base sequence of DNA determines the amino acid sequence of a protein. The chemical and physical properties of each protein are determined by its amino acid sequence, so a single amino acid change is capable of altering the activity of, or even completely inactivating, a protein. **Genotype** refers to the genetic composition

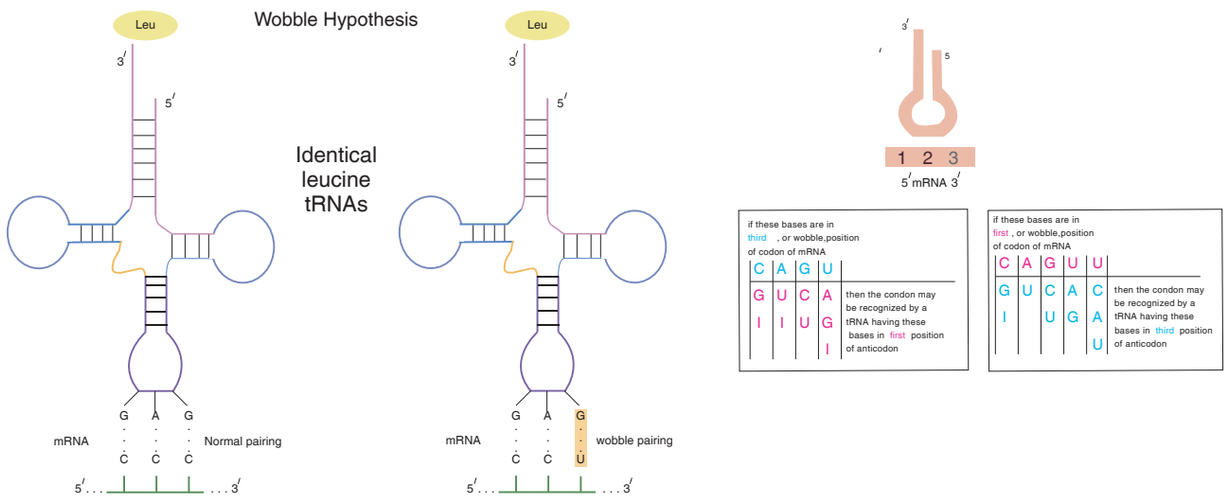


Figure 12.6: Wobble hypothesis

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